

We want to prove $\sum_i g(a_i) \leq \ell$ and, to this aim, it is enough to let $M \rightarrow \infty$. Actually, $\mu^s(\Omega) = \sum_i (a_i - a_i^M) \leq C/M \rightarrow 0$ implies that for each i we have $a_i - a_i^M \rightarrow 0$ and thus $a_i^M \rightarrow a_i$. Using the semi-continuity of g we have $g(a_i) \leq \liminf_{M \rightarrow \infty} g(a_i^M)$. If we fix an arbitrary number N we get

$$\sum_{i=1}^N g(a_i) \leq \liminf_{M \rightarrow \infty} \sum_{i=1}^N g(a_i^M) \leq \ell.$$

By passing to the supremum over N we finally get

$$\sum_{i=1}^{\infty} g(a_i) \leq \ell,$$

which is the claim. $\square$

As a particular example for the functionals of type $\mathcal{F}$ we can consider the L^p norms to the power p , i.e.

$$\mathcal{F}(\mu) = \|\mu\|_{L^p}^p = \begin{cases} \int \rho(x)^p dx & \text{if } \mu = \rho \cdot dx, \\ +\infty & \text{otherwise.} \end{cases}$$

For the functionals of type $\mathcal{G}$, we can instead consider the cardinality of the support, obtained for $g(t) = 1$ if $t > 0$ and $g(0) = 0$:

$$\mathcal{G}(\mu) = \#(\text{spt}(\mu)).$$

Box 7.1. – Good to know! – Characterization of local l.s.c. functionals on $\mathcal{M}^d(\Omega)$

It is also useful to note that functionals of the form $\mathcal{F}$ and $\mathcal{G}$ could be mixed, obtaining local functionals accepting both absolutely continuous and atomic parts. There is a general lower semi-continuity result that comes from the general theory developed in [68, 69, 70], which also characterizes the semi-continuity. It also covers the case of vector-valued measures and can be expressed as follows.

Theorem - Let $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex and l.s.c. and $g : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be subadditive and l.s.c. For every vector $\mathbf{v} \in \mathbb{R}^d$ define

$$f^\infty(\mathbf{v}) := \lim_{t \rightarrow +\infty} \frac{f(t\mathbf{v})}{t} \quad \text{and} \quad g^0(\mathbf{v}) := \lim_{t \rightarrow 0} \frac{g(t\mathbf{v})}{t},$$

and suppose $f^\infty = g^0$. Given a finite positive measure λ , decompose every vector measure $\mu \in \mathcal{M}^d(\Omega)$ as $\mu = \rho \cdot \lambda + \mu^c + \mu^\#$, where $\mu^c + \mu^\#$ is the singular part of μ w.r.t. λ , which is also decomposed into a purely atomic part $\mu^\# = \sum_i a_i \delta_{x_i}$ and the remainder, usually referred to as the Cantor part; $\rho : \Omega \rightarrow \mathbb{R}^d$ is a vector density and the $a_i = \mu(\{x_i\}) \in \mathbb{R}^d$ are also vectors; we also denote by $\mathbf{w} : \Omega \rightarrow S^{d-1}$ the density of μ^c w.r.t. its own total variation measure $|\mu^c|$. Then the functional

$$\mu \mapsto \int f(\rho(x)) d\lambda(x) + \int f^\infty(\mathbf{w}(x)) d|\mu^c|(x) + \sum_i g(a_i)$$

is local and lower semi-continuous for the weak convergence of measures.

Conversely, every local and lower semi-continuous functional on $\mathcal{M}^d(\Omega)$ can be written in the form above, for suitable choices of f , g and λ .

7.2 Convexity, first variations and subdifferentials

We pass in this section to another important notion concerning these functionals. If lower semi-continuity is crucial to establish existence results for minimization problems, convexity properties (in particular, strict convexity) are crucial for uniqueness. Also, with convexity comes the notion of subdifferential, thus we come to another very natural question in calculus of variations: how to compute first variations of these functionals?

We start from the very first question: which among these functionals are convex?

Convexity and strict convexity

- $\mathcal{V}$ is linear and hence convex but not strictly convex.
- $\mathcal{W}$ is quadratic, but not always convex. Take for instance $W(x, y) = |x - y|^2$ and compute

$$\begin{aligned} & \int_{\Omega} \int_{\Omega} |x - y|^2 d\mu(x) d\mu(y) \\ &= \int_{\Omega} \int_{\Omega} |x|^2 d\mu(x) d\mu(y) + \int_{\Omega} \int_{\Omega} |y|^2 d\mu(x) d\mu(y) - 2 \int_{\Omega} \int_{\Omega} x \cdot y d\mu(x) d\mu(y) \\ &= 2 \int_{\Omega} |x|^2 d\mu(x) - 2 \left(\int_{\Omega} x d\mu(x) \right)^2. \end{aligned}$$

This shows that $\mathcal{W}$ has (in this case) a linear part to which we subtract the square of another linear one; it is then concave rather than convex.

- All transport costs can be expressed by duality formula as a supremum of linear functionals

$$\mathcal{T}_c(\mu, \nu) = \sup_{\varphi(x) + \psi(y) \leq c(x, y)} \int \varphi d\mu + \int \psi d\nu$$

and it is hence convex (but W_p is in general not). Strict convexity is discussed later (it is true for $c(x, y) = |x - y|^p$, $p > 1$ and $\nu \ll \mathcal{L}^d$).

- The norm in a dual functional space is always convex since it is a norm, but is never strictly convex because it is 1-homogeneous. Note that also the square of a norm could be non-strictly convex (as it is the case for the L^∞ or the L^1 norms).
- The functionals $\mathcal{F}$ that we considered above are actually convex due to the assumptions on f . Strict convexity is true if f is strictly convex and $L = +\infty$ (for instance if one takes $f(t) = \sqrt{1+t^2} - 1$, which is strictly convex, then $\mathcal{F}$ is not strictly convex because it is finite and linear on singular measures).
- On the contrary, the functionals $\mathcal{G}$ that we considered above are typically not convex since the typical examples of sub-additive functions are concave.

First variations We now pass to the computation of first variations of these functionals. Since many of them are only considered in the convex set $\mathcal{P}(\Omega)$, which is a proper (but convex) subset of the Banach space $\mathcal{M}(\Omega)$, we prefer to give an ad-hoc, and pragmatic, definition.

Definition 7.12. Given a functional $F : \mathcal{P}(\Omega) \rightarrow \mathbb{R} \cup \{+\infty\}$ we say that $\rho \in \mathcal{P}(\Omega)$ is *regular* for F if $F((1-\varepsilon)\rho + \varepsilon\tilde{\rho}) < +\infty$ for every $\varepsilon \in [0, 1]$ and every $\tilde{\rho} \in \mathcal{P}(\Omega) \cap L_c^\infty(\Omega)$ (absolutely continuous probabilities with L^∞ density and compact support), i.e. if the convex envelop of $\{\rho\}$ and $L_c^\infty(\Omega) \cap \mathcal{P}(\Omega)$ is contained in $\{F < +\infty\}$.

If ρ is regular for F , we call $\frac{\delta F}{\delta \rho}(\rho)$, if it exists, any measurable function such that

$$\frac{d}{d\varepsilon} F(\rho + \varepsilon\chi)|_{\varepsilon=0} = \int \frac{\delta F}{\delta \rho}(\rho) d\chi$$

for every perturbation $\chi = \tilde{\rho} - \rho$ with $\tilde{\rho} \in L_c^\infty(\Omega) \cap \mathcal{P}(\Omega)$.

Remark 7.13. From the fact that we necessarily have $\int d\chi = 0$, it is clear that $\frac{\delta F}{\delta \rho}(\rho)$ is defined up to additive constants. On the other hand, up to this invariance, we have uniqueness.

- From $\mathcal{V}(\mu + \varepsilon\chi) = \mathcal{V}(\mu) + \varepsilon \int V d\chi$ we infer $\frac{\delta \mathcal{V}}{\delta \rho}(\rho) = V$.
- As $\mathcal{W}$ is quadratic, the computation is easy:

$$\begin{aligned} \mathcal{W}(\mu + \varepsilon\chi) - \mathcal{W}(\mu) &= \\ \varepsilon \int \int W(x, y) d\mu(x) d\chi(y) + \varepsilon \int \int W(x, y) d\mu(y) d\chi(x) \\ &\quad + \varepsilon^2 \int \int W(x, y) d\chi(x) d\chi(y). \end{aligned}$$

This provides

$$\frac{\delta \mathcal{W}}{\delta \rho}(\rho)(y) = \int W(x, y) d\mu(x) + \int W(y, y') d\mu(y').$$

The formula becomes simpler when W is symmetric (since the two terms are equal) and even simpler when $W(x, y) = h(x - y)$ for an even function h , in which case it takes the form of a convolution $\frac{\delta \mathcal{W}}{\delta \rho}(\rho) = 2h * \mu$.

- For $\mu \mapsto \mathcal{T}_c(\mu, \nu)$, the first variation is given by the Kantorovich potential ϕ , in the transport from μ to ν , provided it is unique; this will be discussed later.
- Analogously, the first variation of $\mu \mapsto \|\mu\|_{\mathcal{X}'}$ is given by the function ϕ realizing the maximum in $\max\{\int \phi d\mu : \|\phi\|_{\mathcal{X}} \leq 1\}$, provided it exists and is unique.
- If $f \in C^1$ and f' satisfies suitable bounds (typically we need f and f' to have polynomial growth), one can see that $\frac{\delta \mathcal{F}}{\delta \rho}(\rho) = f'(\rho)$. Indeed, we can take $\chi =$

$\theta \cdot \lambda$, write $\mathcal{F}(\rho + \varepsilon\chi) = \int f(\rho(x) + \varepsilon\theta(x)) d\lambda(x)$ and differentiate under the integral sign (this requires some growth conditions): we get

$$\frac{d}{d\varepsilon} \mathcal{F}(\rho + \varepsilon\chi)|_{\varepsilon=0} = \int f'(\rho(x)) \theta(x) d\lambda = \int f'(\rho) d\chi.$$

The bounds on f' are only needed to allow derivation under the integral sign.

- Note that in general the functionals $\mathcal{G}$ are not differentiable along this kind of perturbations and that in general no measure is regular for $\mathcal{G}$. Indeed, from $g'(0) = +\infty$ we can infer that $G(\mu + \varepsilon\chi)$ is usually differentiable only if χ is concentrated on the same atoms as μ , even if for χ finitely atomic one would have $\mathcal{G}(\mu + \varepsilon\chi) < +\infty$.

Remark 7.14. We stress the fact that the special role given to L_c^∞ functions in the definition of first variation is completely arbitrary. However, they represent a class of densities which is at the same time dense enough in the class of probabilities and regular enough to guarantee finiteness of most interesting functionals. In particular, we can observe that, for all the functionals above except $\mathcal{G}$, a probability ρ is regular if and only if the value of the functional is finite.

7.2.1 Dual norms

We start from the non-trivial issues above, i.e. the case of the transport costs and of the dual norms. Both are characterized by their definition as sup of linear functionals. Since they are convex functionals, we start from a general fact in convex analysis, which allows us to provide not only the first variation, but also the subdifferential.

Lemma 7.15. *Let $\mathcal{X}$ be a Banach space and $H : \mathcal{X} \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex and l.s.c. Set $H^*(y) = \sup \langle x, y \rangle - H(x)$ for $y \in \mathcal{X}'$, where $\langle x, y \rangle$ denotes the duality between $\mathcal{X}$ and $\mathcal{X}'$. Then $\partial H^*(y_0) = \operatorname{argmax}_x \{ \langle x, y_0 \rangle - H(x) \}$.*

Proof. We know $x_0 \in \partial H^*(y_0) \iff y_0 \in \partial H(x_0)$ (see Box 1.12 in Section 1.6). This is equivalent to the fact that 0 belongs to the subdifferential of $x \mapsto H(x) - \langle x, y_0 \rangle$ at x_0 , but this is also equivalent to the fact that x_0 minimizes the same expression. $\square \quad \square$

Proposition 7.16. *Suppose that $\mathcal{X}$ is a reflexive separable Banach space of functions on Ω such that $\mathcal{X} \cap C_b(\Omega)$ is dense in $\mathcal{X}$. Let $F(\mu) := \|\mu\|_{\mathcal{X}'} = \sup \{ \int \varphi d\mu : \|\varphi\|_{\mathcal{X}} \leq 1 \}$. Suppose that, for a given μ , the function $\varphi_\mu \in \mathcal{X}$ realizing the maximum in the definition of $F(\mu)$ exists and is unique. Then we have $\frac{\delta F}{\delta \rho}(\mu) = \varphi_\mu$. Moreover, the subdifferential $\partial F(\mu)$ (where we consider F to be a functional on the Banach space $\mathcal{M}(\Omega)$, set to $+\infty$ outside $\mathcal{P}(\Omega) \cap \mathcal{X}'$) is always equal to the set (not necessarily a singleton) of maximizers.*

Proof. The second part of the statement is an easy consequence of Lemma 7.15, applied to the functional F defined as $F(\varphi) = 0$ for $\|\varphi\| \leq 1$ and $F(\varphi) = +\infty$ for $\|\varphi\| >$

1. For the first part, take a perturbation χ such that $\mu + \varepsilon\chi$ belongs to $\mathcal{X}'$ for small ε . This implies that both μ and χ belong to $\mathcal{X}'$. From the definition of F , we can write

$$\int \varphi_\mu d\chi \leq \frac{F(\mu + \varepsilon\chi) - F(\mu)}{\varepsilon} \leq \int \varphi_{\mu + \varepsilon\chi} d\chi.$$

In order to pass to the limit, we just need to show that the right hand side tends to $\int \varphi_\mu d\chi$ as $\varepsilon \rightarrow 0$. First, note that the functions $\varphi_{\mu + \varepsilon\chi}$ belong to the unit ball of a reflexive separable Banach space. This means that, up to subsequences, we have $\varphi_{\mu + \varepsilon\chi} \rightharpoonup \varphi$, for a function φ with $\|\varphi\| \leq 1$. We just need to prove $\varphi = \varphi_\mu$. From the uniqueness of the optimizer φ_μ , we just need to show that φ is also an optimizer for μ , i.e. $\int \varphi d\mu = F(\mu)$.

To do this, note that $\mu + \varepsilon\chi$ converges strongly in $\mathcal{X}'$ to μ , and hence we have

$$\lim_{\varepsilon \rightarrow 0} F(\mu + \varepsilon\chi) = \lim_{\varepsilon \rightarrow 0} \int \varphi_{\mu + \varepsilon\chi} d(\mu + \varepsilon\chi) \rightarrow \int \varphi d\mu \leq F(\mu) = \lim_{\varepsilon \rightarrow 0} F(\mu + \varepsilon\chi),$$

where the last equality comes from the continuity of F for strong convergence in $\mathcal{X}'$. Hence the inequality $\int \varphi d\mu \leq F(\mu)$ is indeed an equality and $\varphi = \varphi_\mu$. $\square$ $\square$

7.2.2 Transport costs

We consider here the case of Wasserstein distances to a fixed measure (in particular functionals of the form $\mu \mapsto W_p^p(\mu, \nu)$) and, more generally, transport costs $\mathcal{T}_c(\mu, \nu) := \min\{\int c(x, y) d\gamma : \gamma \in \Pi(\mu, \nu)\}$, i.e. minimal transport costs between measures and a fixed one. Note that for continuity issues it was useful to take advantage of the distance structure of W_p , but for convexity and first variation this will not be the good approach.

For the sake of simplicity, the following result will only be given in the case of a compact domain $\Omega \subset \mathbb{R}^d$.

Proposition 7.17. *Let $\Omega \subset \mathbb{R}^d$ be compact and $c : \Omega \times \Omega \rightarrow \mathbb{R}$ be continuous. Then the functional $\mu \mapsto \mathcal{T}_c(\mu, \nu)$ is convex, and its subdifferential at μ_0 coincides with the set of Kantorovich potentials $\{\varphi \in C^0(\Omega) : \int \varphi d\mu_0 + \int \varphi^c d\nu = \mathcal{T}_c(\mu_0, \nu)\}$. Moreover, if there is a unique c -concave Kantorovich potential from μ_0 to ν up to additive constants, i.e. if the c -concave functions in the above set are all given by the same function φ_μ plus arbitrary additive constants, then we also have $\frac{\delta \mathcal{T}_c(\cdot, \nu)}{\delta \mu}(\mu) = \varphi_\mu$.*

Proof. Let us come back to the expression

$$\mathcal{T}_c(\mu, \nu) = \sup \left\{ \int \varphi d\mu + \int \varphi^c d\nu : \varphi \in C^0(\Omega) \right\},$$

which allows to see $\mathcal{T}_c(\cdot, \nu)$ as a supremum of affine functionals of μ . Here ν is a fixed probability. Note that the very same supremum gives $+\infty$ if we take $\mu \in \mathcal{M}(\Omega) \setminus \mathcal{P}(\Omega)$, i.e.

$$\sup \left\{ \int \varphi d\mu + \int \varphi^c d\nu : \varphi \in C^0(\Omega) \right\} = \begin{cases} \mathcal{T}_c(\mu, \nu) & \text{if } \mu \in \mathcal{P}(\Omega), \\ +\infty & \text{if } \mu \in \mathcal{M}(\Omega) \setminus \mathcal{P}(\Omega). \end{cases}$$

Indeed, if μ is not a positive measure then there exists a function $\varphi \in C^0(\Omega)$, with $\varphi \leq 0$ and $\int \varphi d\mu > 0$. Since $\varphi^c(y) = \inf_x c(x, y) - \varphi(x) \geq \inf c$ (suppose c bounded from below), then $\int \varphi d\mu + \int \varphi^c dv \geq \int \varphi d\mu + \inf c$. By replacing φ with $\lambda \varphi$ and letting $\lambda \rightarrow +\infty$ we see that the supremum is $+\infty$.

On the other hand, if $\mu \geq 0$ but $\mu(\Omega) \neq 1 = v(\Omega)$, we also find $+\infty$ in the sup. Indeed, we can always add a constant λ to any φ , and $(\varphi + \lambda)^c = \varphi^c - \lambda$. Taking for instance $\varphi = 0$ we get

$$\sup \left\{ \int \varphi d\mu + \int \varphi^c dv : \varphi \in C^0(\Omega) \right\} \geq \int 0^c(y) dv(y) + \lambda (\mu(\Omega) - v(\Omega)),$$

which can be made as large as we want if $\mu(\Omega) - v(\Omega) \neq 0$ (here 0^c is a bounded function whose precise expression depends on c).

In order to identify the subdifferential, we apply Lemma 7.15, with $\mathcal{X} = C^0(\Omega)$ (endowed with the sup norm) and $H : \mathcal{X} \rightarrow \mathbb{R}$ given by $H(\varphi) := -\int \varphi^c dv$. We just need to see that this functional is convex and semi-continuous (indeed, it is continuous). Note that if we take $\varphi_0, \varphi_1 \in C^0(\Omega)$ we have

$$\varphi_1^c(y) = \inf_x c(x, y) - \varphi_1(x) \leq \inf_x c(x, y) - \varphi_0(x) + \|\varphi_1 - \varphi_0\|_\infty = \varphi_0^c(y) + \|\varphi_1 - \varphi_0\|_\infty.$$

By interchanging the roles of φ_0 and φ_1 and using the arbitrariness of y we get $\|\varphi_1^c - \varphi_0^c\|_\infty \leq \|\varphi_1 - \varphi_0\|_\infty$, which implies $|\int \varphi_1^c dv - \int \varphi_0^c dv| \leq \|\varphi_1 - \varphi_0\|_\infty$ and hence the continuity of the functional. As far as convexity is concerned, set $\varphi_t = (1-t)\varphi_0 + t\varphi_1$; we have

$$\varphi_t^c(y) = \inf_x c(x, y) - (1-t)\varphi_0(x) - t\varphi_1(x) \geq (1-t)\varphi_0^c(y) + t\varphi_1^c(y).$$

This implies the concavity $\int \varphi_t^c dv \geq (1-t) \int \varphi_0^c dv + t \int \varphi_1^c dv$ and hence H is convex.

In order to get the first variation in the situation described in the statement (uniqueness of the c -concave Kantorovich potential), we take $\mu_\varepsilon = \mu + \varepsilon\chi$ with $\chi = \tilde{\mu} - \mu$ and we estimate the ratio $(\mathcal{T}_c(\mu_\varepsilon, \nu) - \mathcal{T}_c(\mu, \nu))/\varepsilon$. First, by using that $(\varphi_\mu, \varphi_\mu^c)$ is optimal in the dual formulation for μ , but not necessarily for μ_ε , we have

$$\frac{\mathcal{T}_c(\mu_\varepsilon, \nu) - \mathcal{T}_c(\mu, \nu)}{\varepsilon} \geq \frac{\int \varphi_\mu d\mu_\varepsilon + \int \varphi_\mu^c dv - \int \varphi_\mu d\mu - \int \varphi_\mu^c dv}{\varepsilon} = \int \varphi_\mu d\chi,$$

which gives the lower bound $\liminf_{\varepsilon \rightarrow 0} (\mathcal{T}_c(\mu_\varepsilon, \nu) - \mathcal{T}_c(\mu, \nu))/\varepsilon \geq \int \varphi_\mu d\chi$.

To look at the limsup, first fix a sequence of values of ε_k realizing the limsup, i.e. $\lim_k (\mathcal{T}_c(\mu_{\varepsilon_k}, \nu) - \mathcal{T}_c(\mu, \nu))/\varepsilon_k = \limsup_{\varepsilon \rightarrow 0} (\mathcal{T}_c(\mu_\varepsilon, \nu) - \mathcal{T}_c(\mu, \nu))/\varepsilon$. Then we can estimate the same ratio using the optimality of a pair (φ_k, φ_k^c) , Kantorovich potentials in the transport from μ_{ε_k} to ν (we can assume them to be c -concave functions and such that $\varphi_k(x_0) = 0$ for a fixed point $x_0 \in \Omega$). Now we may write

$$\frac{\mathcal{T}_c(\mu_{\varepsilon_k}, \nu) - \mathcal{T}_c(\mu, \nu)}{\varepsilon_k} \leq \frac{\int \varphi_k d\mu_{\varepsilon_k} + \int \varphi_k^c dv - \int \varphi_k d\mu - \int \varphi_k^c dv}{\varepsilon_k} = \int \varphi_k d\chi. \quad (7.1)$$

As in Proposition 7.16, we need to pass to the limit in k . Theorem 1.52 shows that we have uniform convergence (up to extracting another subsequence) $(\varphi_k, \varphi_k^c) \rightarrow (\varphi, \varphi^c)$

and that (φ, φ^c) must be optimal in the duality formula for the transport between μ and ν . This implies $\varphi = \varphi_\mu$ by uniqueness. Finally, passing to the limit in (7.1) we get also $\limsup_{\varepsilon \rightarrow 0} (\mathcal{T}_c(\mu_\varepsilon, \nu) - \mathcal{T}_c(\mu, \nu)) / \varepsilon \leq \int \varphi_\mu d\chi$. $\square$ $\square$

As Proposition 7.17 requires uniqueness of the Kantorovich potential in order to efficiently compute the first variation, we also give here a sufficient condition

Proposition 7.18. *If Ω is the closure of a bounded connected open set, c is C^1 , and at least one of the measures μ or ν is supported on the whole Ω , then the c -concave Kantorovich potential in the transport from μ to ν is unique up to additive constants.*

Proof. Suppose that $\text{spt}(\mu) = \Omega$. First note that $c \in C^1$ implies that c is Lipschitz on $\Omega \times \Omega$, and hence all Kantorovich potentials, which are c -concave, are Lipschitz as well, and are differentiable a.e. Consider two different Kantorovich potentials φ_0 and φ_1 . We use Proposition 1.15, which guarantees that their gradients must agree a.e. on Ω . Since Ω is the closure of a connected open set, this means that the difference $\varphi_0 - \varphi_1$ is constant and provides the desired result. If the measure with full support is ν , just apply the same procedure to the transport from ν to μ and get the uniqueness of the potential ψ . Then, from $\varphi = \psi^c$ (here we use c -concavity) one also recovers the uniqueness of φ . $\square$ $\square$

Note that in order to apply the above result to Proposition 7.17, the uniqueness μ -a.e. would not be enough (since one integrates it also against other measures μ_ε), and, anyway, without connectedness assumptions on $\text{spt}(\mu)$ it would be impossible to deduce the uniqueness of φ .

We finish this section with a remark on strict convexity. We come back for simplicity to the case $c(x, y) = |x - y|^p$, i.e. to the functional $W_p^p(\cdot, \nu)$, but the reader will see that everything works the same under the twist condition (or if $c(x, y) = h(x - y)$ with h strictly convex). Also, the assumption $\nu \ll \mathcal{L}^d$ is not sharp, at least for $p = 2$ (see Section 1.3.1).

Proposition 7.19. *If $\nu \ll \mathcal{L}^d$ and $p > 1$ the functional $W_p^p(\cdot, \nu)$ is strictly convex.*

Proof. Suppose by contradiction that $\mu_0 \neq \mu_1$ and $t \in]0, 1[$ are such that $W_p^p(\mu_t, \nu) = (1 - t)W_p^p(\mu_0, \nu) + tW_p^p(\mu_1, \nu)$, where $\mu_t = (1 - t)\mu_0 + t\mu_1$. Let γ_0 be the optimal transport plan in the transport from ν to μ_0 (pay attention to the direction: it is a transport map if we see it backward: from ν to μ_0 , since $\nu \ll \mathcal{L}^d$); we write $\gamma_0 = (T_0, \text{id})_\# \nu$. Analogously, take $\gamma_1 = (T_1, \text{id})_\# \nu$ optimal from ν to μ_1 . Set $\gamma_t := (1 - t)\gamma_0 + t\gamma_1 \in \Pi(\mu_t, \nu)$. We have

$$\begin{aligned} (1 - t)W_p^p(\mu_0, \nu) + tW_p^p(\mu_1, \nu) &= W_p^p(\mu_t, \nu) \leq \int |x - y|^p d\gamma_t \\ &= (1 - t)W_p^p(\mu_0, \nu) + tW_p^p(\mu_1, \nu), \end{aligned}$$

which implies that γ_t is actually optimal in the transport from ν to μ_t . Yet γ_t is not induced from a transport map, unless $T_0 = T_1$. This is a contradiction with $\mu_0 \neq \mu_1$ and proves strict convexity. $\square$ $\square$

7.2.3 Optimality conditions

We finish this part of the chapter devoted to first variations with an important computation, which leads to optimality conditions in the space $\mathcal{P}(\Omega)$. We do not claim this result to be sharp, but it is enough in most situations, and the reader can easily see how to adapt it to other cases.

Proposition 7.20. *Suppose that ρ_0 minimizes a functional $F : \mathcal{P}(\Omega) \rightarrow \mathbb{R} \cup \{+\infty\}$, that ρ_0 is regular for F (see Definition 7.12) and that there exists $\frac{\delta F}{\delta \rho}(\rho_0)$. Call this function g and set $\ell = \text{ess inf } g$, the essential infimum of g .*

First case : Suppose that $g \in C_0(\Omega)$. Then $\text{spt}(\rho_0) \subset \text{argmin } g = \{g = \ell\}$, and we have everywhere the inequality $g \geq \ell$, with equality $g = \ell$ on $\text{spt}(\rho_0)$.

Second case : Suppose that g is only measurable, but $\rho_0 \ll \mathcal{L}^d$. Then we have a.e. the inequality $g \geq \ell$, with $g = \ell$ a.e. on $\{\rho_0 > 0\}$.

Proof. Consider a competitor $\tilde{\rho} \in L_c^\infty(\Omega)$. Take $\rho_\varepsilon := (1 - \varepsilon)\rho + \varepsilon\tilde{\rho} = \rho + \varepsilon\chi$, with $\chi = \tilde{\rho} - \rho$. The measure ρ_ε is an admissible competitor in the minimization problem as soon as $\varepsilon \in [0, 1]$. We deduce that

$$\frac{d}{d\varepsilon}(F(\rho_\varepsilon))|_{\varepsilon=0} \geq 0.$$

This means

$$\int g d\chi \geq 0, \quad \text{i.e.} \quad \int g d\tilde{\rho} \geq \int g d\rho_0.$$

We now use the arbitrariness of $\tilde{\rho}$. Take any $\ell' > \ell$. By definition of essential infimum the set $\{g < \ell'\}$ has positive Lebesgue measure, and we can choose an L^∞ density $\tilde{\rho}$ concentrated on it. We get $\ell' > \int g d\rho_0$. Letting $\ell' \rightarrow \ell$, we get $\ell \geq \int g d\rho_0$.

In the first case, the essential infimum coincides with the infimum, and hence we have $g \geq \ell$ everywhere. But integrating this inequality against ρ_0 we have the opposite inequality, which implies $g = \ell$ ρ_0 -a.e. This equality extends to the support of ρ_0 since g is continuous.

In the second case, we have $g \geq \ell$ a.e. Since ρ_0 is supposed to be absolutely continuous, the same is true ρ_0 -a.e., and hence we have $g = \ell$ ρ_0 -a.e., which means a.e. on $\{\rho_0 > 0\}$. □

We will see for instance in Sections 7.4.1 and 8.3 some applications of this criterion.

We also stress that, when F is convex, alternative arguments using subdifferentials could be used, but the above strategy is the only possible one when we do not have convexity. We can see how it works in some examples.

Example 7.21. Consider

$$\min \left\{ F(\rho) := W_2^2(\rho, \nu) + \int V(x) d\rho(x) : \rho \in \mathcal{P}(\Omega) \right\},$$

where $\Omega \subset \mathbb{R}^d$ is compact and V continuous. In such a case it is clear that a minimizer ρ_0 exists and that it is regular for F since F is finite on the whole space $\mathcal{P}(\Omega)$. If the

Kantorovich potential φ (for the cost $c(x, y) = |x - y|^2$) in the transport from ρ_0 to ν is unique up to additive constants, then the first variation of F at ρ_0 is given by $\varphi + V$, and we can apply Proposition 7.20 to obtain the existence of a constant C such that

$$\varphi + V \geq C \text{ on } \Omega, \quad \varphi + V = C \text{ on } \text{spt}(\rho_0).$$

This is the case if $\text{spt}(\nu) = \Omega$, for instance.

Another way to obtain the optimality conditions is the following: the second term of F is a continuous and linear functional on $\mathcal{M}(\Omega)$, and the first is a convex functional defined on $\mathcal{M}(\Omega)$, with subdifferential given by the set of Kantorovich potentials. Hence, we can prove

$$\partial F(\rho_0) = \{V + \varphi : \varphi \text{ Kantorovich potential from } \rho_0 \text{ to } \nu\}.$$

From standard optimality conditions in convex analysis we obtain the existence of a potential φ such that $V + \varphi = 0$. In this case the result that we get is much stronger with this convex procedure, since we get rid of the inequality case. Note, however, that the function φ that we get is not necessarily c -concave: since we know $\varphi = \varphi^{cc}$ on $\text{spt}(\rho_0)$, this means that it has been modified out of $\text{spt}(\rho_0)$ in order to guarantee the equality.

Example 7.22. Consider

$$\min \left\{ F(\rho) := W_2^2(\rho, \nu) + \int_{\Omega} \frac{1}{p} \rho^p(x) dx : \rho \in \mathcal{P}(\Omega), \rho \ll \mathcal{L}^d \right\},$$

where $\Omega \subset \mathbb{R}^d$ is compact and $p > 1$. Again, a minimizer $\rho_0 \in L^p(\Omega)$ exists and is regular (in the sense of Definition 7.12), since F is finite on $L^p(\Omega) \cap \mathcal{P}(\Omega)$ which includes for sure all convex combinations of ρ_0 and L^∞ functions. Computing the first variation and using Proposition 7.20, we obtain the existence of a constant C such that

$$\varphi + \rho^{p-1} \geq C \text{ on } \Omega, \quad \varphi + \rho^{p-1} = C \text{ on } \text{spt}(\rho_0).$$

This, by the way, can be re-written as $\rho^{p-1} = (C - \varphi)_+$.

On the other hand, even if the functional is still convex, it is not easy to compute the subdifferential of F , viewed as a functional on $\mathcal{M}(\Omega)$ (set to $+\infty$ outside L^p probabilities). Indeed, the subdifferential of a sum is not always the sum of the subdifferentials (see for instance [158] for convex analysis issues in infinite dimension; however, this requires continuity of at least one of the two functionals, while here both are finite on a small subset of $\mathcal{M}(\Omega)$). Moreover the first variation of the second term, i.e. ρ^{p-1} , does not belong in general to the space $C(\Omega)$, i.e. to the space $\mathcal{M}(\Omega)$ is in duality with. A way of circumventing this difficulty would be to view F as a functional on $L^p(\Omega)$. Here, the second term becomes a continuous convex functional. The first one must be re-written as $\sup\{\int \phi d\rho + \int \phi^c d\nu : \phi \in L^{p'}(\Omega)\}$ (with a proper definition of the c -transform in terms of essential infima instead of infima). This would provide the optimality condition: there exists an $L^{p'}$ Kantorovich potential φ such that $\varphi + \rho^{p-1} = 0$. Again, $\varphi = \varphi^{cc}$ ρ_0 -a.e., and $\varphi \leq \varphi^{cc}$ elsewhere. This gives again

$$\varphi^{cc} + \rho^{p-1} \geq C \text{ on } \Omega, \quad \varphi^{cc} + \rho^{p-1} = C \text{ on } \text{spt}(\rho_0).$$

7.3 Displacement convexity

In all the considerations of the previous section about convexity, we always considered the standard convex interpolations $[0, 1] \ni t \mapsto (1-t)\mu_0 + t\mu_1$. Yet, another notion of convexity, more linked to the metric structure of the space $\mathbb{W}_p(\Omega)$, may be useful.

Let us start from this general definition.

Definition 7.23. In a geodesic metric space X , we define $F : X \rightarrow \mathbb{R} \cup \{+\infty\}$ to be *geodesically convex* if for every two points $x_0, x_1 \in X$ there exists a constant speed geodesic ω connecting $\omega(0) = x_0$ to $\omega(1) = x_1$ such that $[0, 1] \ni t \mapsto F(\omega(t))$ is convex.

This obviously reduces to usual convexity in $\mathbb{R}^d$ or in any other normed vector space where segments are the unique geodesics (note that in any normed vector space segments are geodesics but they may not be the unique ones, as it is the case in L^1 or L^∞ or any non-strictly convex space). By the way, in spaces where there is not uniqueness of the geodesics the definition we gave could be discussed, since one could also choose to define as geodesically convex those functionals that satisfy the geodesic inequality on *every* geodesic ω . . . But the one we gave is the definition that is usually chosen, as it satisfies some extra stability results that would not be true with the more restrictive one requiring convexity for every geodesic.

The notion of geodesic convexity in the space $\mathbb{W}_p(\Omega)$ has been introduced by McCann in [229], and is particularly interesting since we know how to characterize geodesics in such a space. This notion of convexity is usually referred to as *displacement convexity*. Note that it could a priori depend on the exponent p .

7.3.1 Displacement convexity of $\mathcal{V}$ and $\mathcal{W}$

We recall that the geodesics for the W_p distance are of the form $\mu_t = (\pi_t)_\# \gamma$ where γ is an optimal transport for $c(x, y) = |x - y|^p$ and $\pi_t(x, y) = (1-t)x + ty$.

Proposition 7.24. *The functional $\mathcal{V}$ is displacement convex if and only if V is convex. The functional $\mathcal{W}$ is displacement convex if W is convex.*

Proof. First consider $\mathcal{V}$ and suppose that V is convex. Let us evaluate $\mathcal{V}(\mu_t)$:

$$\mathcal{V}(\mu_t) = \int V d(\pi_t)_\# \gamma = \int V((1-t)x + ty) d\gamma.$$

It is clear from this formula that $t \mapsto \mathcal{V}(\mu_t)$ is convex if V is convex.

On the other hand, the convexity of V is a necessary condition for $\mathcal{V}$ being convex as one can easily check by considering geodesics of the form $\mu_t = \delta_{(1-t)x + ty}$ since $\mathcal{V}(\mu_t) = V((1-t)x + ty)$.

The proof for the convexity of $\mathcal{W}$ is similar: consider

$$\begin{aligned} \mathcal{W}(\mu_t) &= \int W(x, x') d(\pi_t)_\# \gamma(x) d(\pi_t)_\# \gamma(x') \\ &= \int W((1-t)x + ty, (1-t)x' + ty') d\gamma(x, y) d\gamma(x', y'), \end{aligned}$$

which easily gives the condition for the convexity. $\square$ $\square$